Syed Ziaul Bin Bashar (He/Him)

Ph.D. Researcher | Artificial Intelligence, Machine Learning & Cyber-Physical Security

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Research Profile

Computer Science Ph.D. researcher working on LLM-driven intelligent systems workflows, computer vision for process monitoring, and cyber-physical security.

Experience

Graduate Research Assistant Aug 2026 – Present
The University of Texas at Dallas | Richardson, TX

- Conducting doctoral research under Dr. Satyajayant Misra.

Graduate Research Assistant | Networked Systems and Optimization Lab (NSOL) Sep 2024 – Aug 2026
New Mexico State University | Las Cruces, NM

- LLM-Driven Pipeline in Additive Manufacturing. 1 publication.
- Real-time manufacturing quality monitoring, top-layer segmentation. 1 pending publication.

Data Analyst | Supply Chain Feb 2023 – Aug 2024
Unilever Bangladesh Limited | Dhaka, Bangladesh

- Machine learning on inventory forecasting; received the Supply Chain Director's Award (H1 2023).
- Analyzed Allobot and worked on a digital twin for daily product allocation to customers.

Education

Doctor of Philosophy - Computer Science Aug 2026 – Present
The University of Texas at Dallas | Richardson, TX
Research advisor: Dr. Satyajayant Misra.

Doctor of Philosophy - Computer Science Aug 2024 – Aug 2026
New Mexico State University | Las Cruces, NM
CGPA: 3.89/4.00 | Research advisor: Dr. Satyajayant Misra. Continued doctoral studies at UT Dallas.

Bachelor of Computer Science & Engineering Jan 2019 – Jan 2023
BRAC University | Dhaka, Bangladesh
CGPA: 3.46/4.00 | Named in the Vice Chancellor's list for 4 Semesters.

Peer-Reviewed Publications and Research Work

AutoSlice: A Fully Automatic LLM-driven Slicing Configuration Framework for Additive Manufacturing
First author | 54th North American Manufacturing Research Conference (NAMRC 54), 2026 | Paper Link

Comparative Study of Object Detection Models for Safety in Autonomous Vehicles, Homes, and Roads Using IoT Devices
Third author | IEEE CSDE, 2023 | Paper Link

IPP-STL: Intellectual Property Protection in Distributed Additive Manufacturing via Surrogate STL Generation and G-code Restoration
First author | Submitted to *Additive Manufacturing Letters*

SoK: A Cyber-Physical Reference Model and Attack Taxonomy for Additive Manufacturing
First author | Manuscript in preparation

G-code Gated U-Net (GG-Net) for Real-time Layer-wise In-situ Quality Control in Additive Manufacturing Processes
Third author | Manuscript in preparation

Projects

Jailbreaking AI Detectors: Tested robustness of different AI detectors under social media style compression.

G-Code Explanation using NLP: Fine-tuning small language models (SLMs) to generate concise natural-language explanations for 3D-printer G-code commands.

GCodeLM: A framework for detecting malicious changes in 3D-printer G-code commands | finetuned lightweight Llama model with G-code-aware tokenization and classification.

Shuffled STL: An algorithm to deploy stealthy layer-level attacks on additive manufacturing pipelines.

Technical Skills

Programming Python, Java, Julia, C++, JavaScript, SQL

AI / ML Large language model (LLM) workflows, machine learning, object detection, image segmentation, inventory forecasting

Software / Data Spring Boot, REST APIs, PostgreSQL, Docker, Git, Power BI, Selenium, Postman, pgAdmin

Manufacturing / Research OctoPrint, PrusaSlicer, AutoCAD, Webots, LaTeX

Volunteer

- **President**, Computer Science Graduate Student Organization, NMSU — Aug 2025 – Aug 2026
- **IMTS 2026**: Student Summit — Sep 2026
  Demonstrated an LLM-driven text-to-print workflow for automated CAD generation and slicing.
- **Judge**, New Mexico TSA State Conference, high-school coding event. — Feb 2026
- Unilever Heart-to-Heart Market; supported fundraising for 300 underserved families. — Mar 2024